Week 1: Intro to Course & Terminology

STA237: Probability, Statistics, and Data Analysis I

Prof. Kagba N. Suaray

Week of Sept. 8, 2026

Welcome to STA237!

This Week's Schedule

1. Course Introductions: Welcome and Outcomes
2. Course Introductions: Tutorials - What to expect?
3. Course Introductions: Study tips!
4. Introduce some key terminologies

Why STA237?

This course covers probability, statistics and data analysis.

POLL 1.1: Why do these topics matter?

Welcome to STA237!

This is an introductory course on probability. Our main focus in this course is to develop an understanding of probability, the concept of probability distributions, and modeling behaviours of random quantities both discrete and continuous. We will work towards understanding how random quantities can be modeled, describing and interpreting their distribution features, and using simulations in R to help visually represent them.

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- Describe the association between two random quantities using bivariate distributions
- Describe the limiting behaviour of sums and averages of random quantities

Course Syllabus

The course syllabus is an official document that outlines the policies and instructions that both students and instructors agree to follow. Students are expected to be familiar with the policies and instructions within, and to refer to it often if/when they have questions about course policies. Note that only a draft version of the course syllabus has been posted. There will be some minor changes before the official course syllabus will be shared.

Course Structure

- **Lecture:** All lecture information and materials can be found on our Quercus site. There are no lecture recordings. You can find lecture information including what is covered that week, blank and annotated (if available) slides on the weekly pages, which you'll find by clicking your respective LEC section button in **In-Class Materials**:

In-Class Materials

Section	Day/Time	Location	Slides
L0101 *Prof. Kagba Suaray*	T 11 am - 1 pm R 10 - 11 am	SF 1105	
L0201 *Prof. Kagba Suaray*	T 1 - 3 pm R 1 - 2 pm	PB B250 HS 610	

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- **Tutorials:** There are weekly mandatory tutorials starting next week (week of Sept. 12). Each tutorial you will either be completing a pair learning activity or writing a quiz. See course syllabus or Quercus Tutorial pages for the specific schedule.

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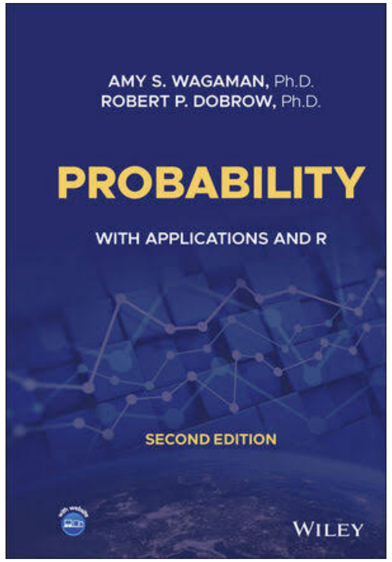
NOTE: There are significant assessment, tutorial, course pacing differences between lecture sections that run on. Tutorial assessments also differ by lecture section. If you write in a tutorial that you are not officially enrolled, your work will not be graded. Because of this, students are **not permitted** to mix and match tutorial and lecture sections.

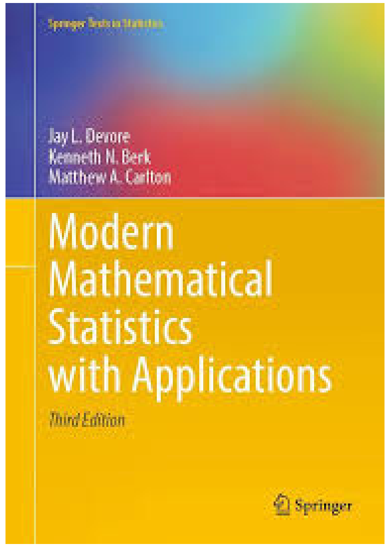
Course Structure

- **Practice Problems/Other Resources:** Suggested practice problems are posted every week on the weekly pages. Relevant textbook sections are also provided for optional, supplemental reading. Students are expected to properly attempt at least half of the suggested problems for regular practice and to help them apply and consolidate concepts.

Our course uses three reference textbooks. They each have their individual strengths:



1. **Probability with Applications and R, 2nd Edition**
   - Fairly approachable to read, many practice problems using R, not as many practice questions that practice similar skills, but diverse questions



2. **Modern Mathematical Statistics with Applications, 3rd Edition**
   - Classic textbook, many and diverse practice questions, good for standard test/quiz prep

3. A Modern Introduction to Probability and Statistics: Understanding Why and How (Optional):
   - Easiest and most approachable to read, diverse and interesting questions

Course Help

- **Where to get course help:** For course-content related questions, please make use of the weekly discussion boards, instructor and TA office hours, and the Statistics Aid Centres. Instructor office hours are not restricted by section!
- **Discussion Boards:** There will be weekly discussion boards for students to pose questions on that week's content, as well as practice problem discussion boards where you can post what you've tried, where you might have gotten stuck on a problem, or just get feedback on how you're presenting your solutions!

Discussion boards are a supportive learning space for everyone and will be monitored by the teaching team. Students are encouraged to participate both in asking and answering questions. Engaging in this process of explaining has been shown to be an *effective way to learn*, consolidate, and retain what you learn. See:

- Why does explaining help learning? Insight from an explanation impairment effect by Princeton University
- Learning by teaching others is extremely effective: Why does explaining help learning? Insight from an explanation impairment effect by the British Psychological Society

Grading Scheme (Tentative)

Syllabus Quiz	0.5%	Available from Sept. 8 to 23
Study Skills Workshop	1%	Synchronous on OR Asynchronous
Pair Activities (Best 5 of 6)	1.5% each	**Weekly during tutorial, see Quercus or Table 2 in syllabus for schedule.**
Quiz (Best 3 of 4)	6% each	
LearnR Modules (4)	2% each	Due: Oct. 9, Oct. 23, Nov. 20, Dec. 7
Midterm	25%	Sat. Nov. 7th, 10:10 - 11:40 am (*)
Final Exam	40%	Scheduled by FAS
BONUS: Pre-/Post- Course Surveys & Midterm Survey	0.5% each	Due: Sept. 22 & Dec. 8 Nov. 17

(*) NOTE: Due to room scheduling, please budget for up to a 15 minute delay in start/end times.

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- Do the same on discussion boards

Volunteer Notetaker Needed In This Class! Sign up today

- We are looking for dedicated students to sign up as a notetaker in this course.
- Support your peers who are registered with Accessibility Services by volunteering as a notetaker.
- Upon completion of volunteering, notetakers are eligible to receive a certificate of appreciation, and add the opportunity to your Co-Curricular Record (CCR)
- Sign up by following the link below or the QR code. For further support please see: How to Become a Volunteer Note-taker - U of T Accessibility Services

For more information, please check out https://uoft.me/notetaking

Questions?

Introduction to Key Terminology

In this course, we will be learning about quantities that occur via a random process, and ways we can model this random behaviour. To do so, we need to learn some key definitions:

Random Experiment

A process that allows us to gather data, observations (or realizations) where we know the set of possible outcomes of the experiment but the outcome of any one trial cannot be known exactly even if we were to repeat the process under the same conditions.

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Sample Space

The set of all possible outcomes of a random experiment. Usually denoted with Ω or S. The elements in a sample space are determined by the **outcome of interest**. Elements in the sample space are denoted by ω.

Examples:

- Rolling a die and the resulting top-facing number
- Rolling a pair of dice and the sum of the top-facing numbers
- Tossing a coin and the time it spent in the air before landing

-
-
-

Event and Operators

Event

A subset of the sample space, usually denoted by a capital letter near the start of the alphabet. A **simple event** is a subset with exactly one element of the sample space, while a **compound event** is a subset containing multiple elements of the sample space. The **complement event** of the event A is the set of outcomes in Ω that are **not** in A. This is often denoted as one of: A^c, A, or A'.

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Consider two events A and B in the sample space Ω.

- The **union** of two events A and B is the event that contains elements that are in A, B, or both. The union of two events is denoted as __________.
    - Union of events is usually described with the word “or”, or “either”. e.g., “A or B”, or “Either A or B”
    - Notice that __________ = Ω

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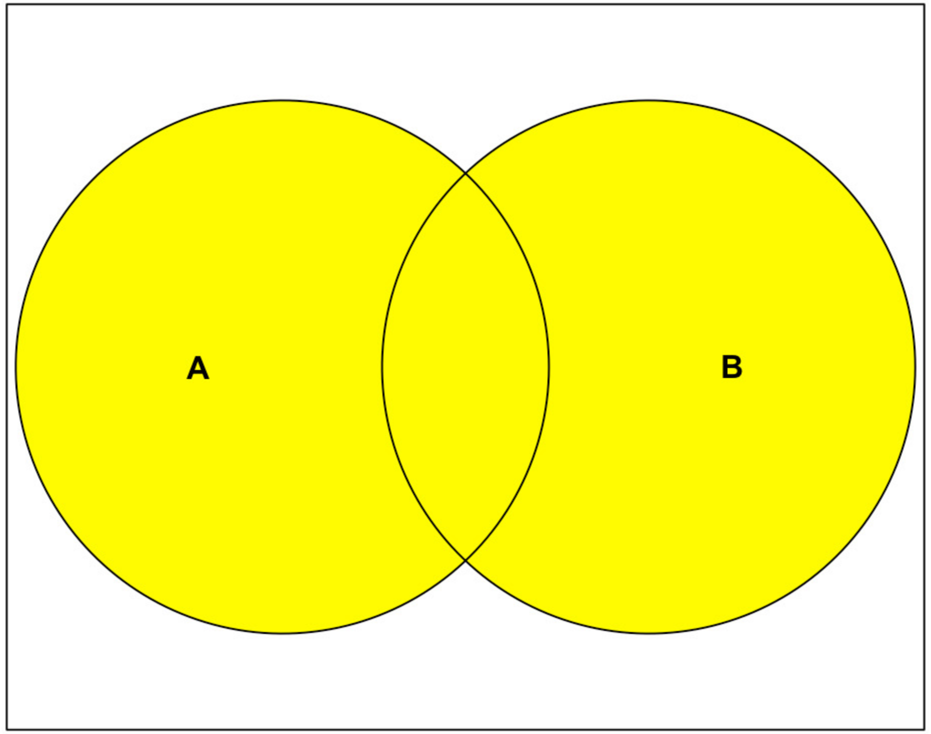
  - Union of events is usually described with the word "or", or "either". e.g., "A or B", or "Either A or B"
  - Notice that ________ = Ω
- The **intersection** of two events A and B is the event that contains elements that are common to *both* A and B. The intersection of two events is commonly denoted as ________ or ________.
  - Intersection of events is usually described with the word "and" or "both". e.g., "A and B", "Both A and B".
  - Notice that ________ = $\varnothing$

Visualizing Events



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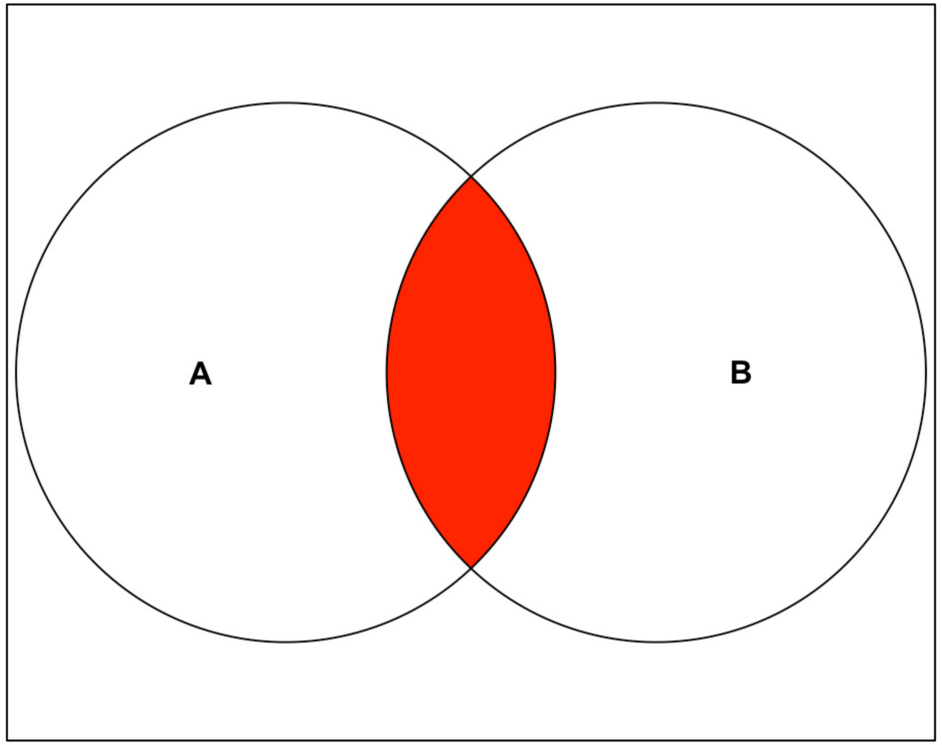

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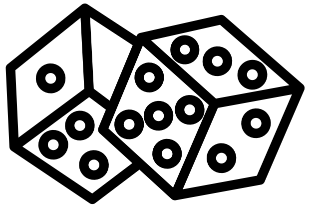
POLL 1.2: Which of these represents the union of A and B ?

Event Operator Examples



Let's keep things simple. Suppose we roll two dice and record the pairs of outcomes. List the following events:

a. The sample space

b. The event that doubles were rolled.

c. The event that the outcome sums to 8.

d. The event where one die has twice the face value of the other.

Event Relations

Consider two events A and B in the sample space Ω:

- Two events are **independent** if the occurrence of one event does not alter the chance that the other event occurs.
- Two events are **dependent** if the occurrence of one event alters the chance that the other event occurs.
- Two events are **mutually exclusive** or **disjoint** if events A and B cannot occur together or simultaneously as an outcome of a random experiment

Independence vs Mutually Exclusive

It's common to mix these two terms up because they *seem* similar!

Independence

Whether event A occurs or not doesn't impact the chances of B occurring.

- If A occurs, then the chance of B is ________
- If A does not occur, then the chance of B is ________

E.g., Let Ω be the sample space for the result of two consecutive coin tosses. The sample space would be $\Omega = \{$____________________$\}$. Let:

- A be the event that the coin landed heads on the **first** toss
- B be the event that the coin landed tails on the **second** toss

Mutually Exclusive

Whether event A occurs or not directly impacts the chances of B occurring

- If A occurs, then the chance of B is ________
- If A does not occur, then the chance of B is ________

E.g., Let Ω be the sample space for the result of two consecutive coin tosses. The sample space would be $\Omega = \{$____________________$\}$. Let:

- A be the event that the coin landed heads on the **first** toss
- B be the event that the coin landed tails on the **first** toss

Mutual exclusivity is a special case of ________________________.

Example 2 - Event Relations

Determine whether the following pairs of events or outcomes are mutually exclusive, independent, or dependent. Opt for the most precise label where two answers are possible.

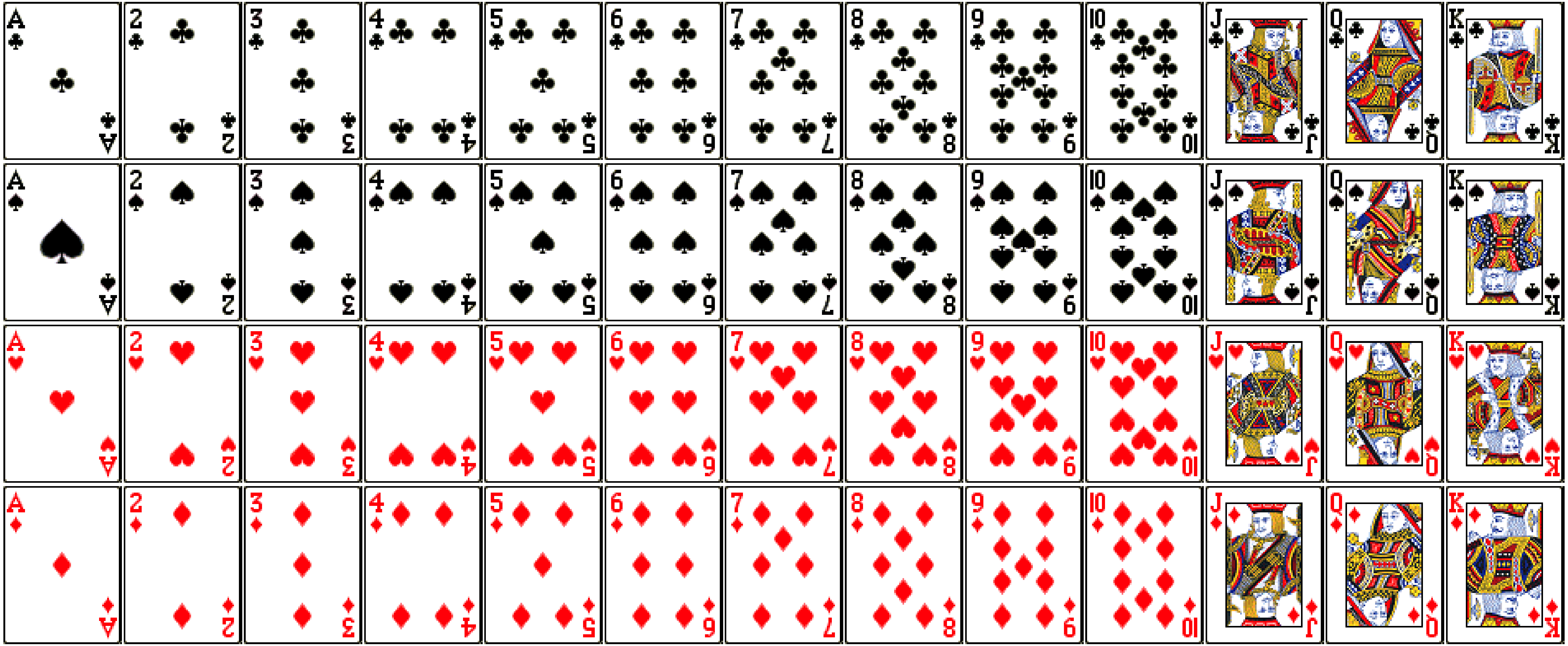
1. A student at U of T is randomly selected and surveyed about their course work. Let $A = \{\text{Enrolled in 100 level course}\}$ and $B = \{\text{Enrolled in 300 level course}\}$

2. A card is drawn at random from a deck of 52 playing cards (see next slide). Let $A = \{\text{Black card}\}$ and $B = \{\text{Ace of Spades}\}$.

3. You and your friend play three games. Let $A = \{\text{You win all WWW}\}$ and $B = \{\text{Your (former) friend wins all LLL}\}$

Traditional Deck of Cards

Keep in Mind for next week's Tutorial!



Commutative, Associative, Distributive Laws

Unions and intersections of events have some useful properties. Consider a sample space Ω with three events A, B, C:

- **Commutative Law**: $A \cup B = B \cup A$
- **Associative Law**: $(A \cup B) \cup C = A \cup (B \cup C)$ and $(A \cap B) \cap C = A \cap (B \cap C)$
- **Distributive Law**: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

Verify these properties using a Venn diagram!

DeMorgan's Laws

For two events A and B:

- $(A \cup B)^c =$
- $(A \cap B)^c =$

This can be extended more generally to the set of events $\{A_1, A_2, \ldots, A_n\}$:

- $(\bigcup_{i=1}^{n} A_i)^c = \bigcap_{i=1}^{n} A_i^c$
- $(\bigcap_{i=1}^{n} A_i)^c = \bigcup_{i=1}^{n} A_i^c$

DeMorgan's Laws can be very helpful in re-expressing event operators into more familiar forms while solving problems or in proofs of event relations.

Probability Axioms

Probability Function

In a random experiment with sample space Ω, a **probability function** P is a function on Ω that assigns to each element ω in the sample space a numerical value that measures the *chance* that event ω will occur. This is denoted as $P(\omega)$. A probability function has the following properties:

1. $$P(\omega) \geq 0$$

   for all $\omega \in \Omega$

2. $$\sum_{\omega \in \Omega} P(\omega) = 1$$

3. For all events $A \subseteq \Omega$,

   $$P(A) = \sum_{\omega \in A} P(\omega)$$

Probability and Event Relations

All probability models have the three properties, or "probability axioms" listed on the previous slide. Using these axioms, we can show how probabilities of events are related!

Probability of Complementary Event

POLL 1.3: Intuitively, for a given event A with probability $P(A)$, we would expect that the probability of its complement A^c will be $P(A^c)$ = __________.

Example 3 - Probability and Event Relations

In a group of 25 U of T students, 12 had eaten a carne asada burrito, 15 had eaten bulgogi, and 6 have neither eaten a carne asada burrito nor bulgogi. Try to represent this information in a Venn diagram.

POLL 1.4: Can you determine how many friends have eaten a carne asada burrito AND bulgogi?

Probability and Event Relations

Example 3 is a classic example of the **Inclusion-Exclusion** principle, sometimes called the Addition Rule, which tells us for two events A and B in a sample space:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Let’s show this using probability axioms:

Example 4 - Inclusion-Exclusion Principle

Among a group of 60 students, 28 could roll their tongue and 15 have attached earlobes. 8 students could both roll their tongue and have attached earlobes. We conduct a random experiment where a student is sampled at random and we note whether they could roll their tongue or if they have attached earlobes. Let

- T be the event that the student can roll their tongue.
- E be the event that the student has attached earlobes.

Represent the following events in notation, and determine the number of elements in each event:

a. The student cannot roll their tongue.
b. The student can neither roll their tongue nor has attached earlobes.

Example 5 - Addition Rule

MMSA Ex. 15 A consulting firm presently has bids out on three projects. Let $A_i = \{Awarded\ project\ i\}$ for $i = 1, 2, 3$ and suppose that:

- $P(A_1) = 0.22$
- $P(A_2) = 0.25$
- $P(A_3) = 0.28$
- $P(A_1 \cap A_2) = 0.11$
- $P(A_1 \cap A_3) = 0.05$
- $P(A_2 \cap A_3) = 0.07$
- $P(A_1 \cap A_2 \cap A_3) = 0.01$

Describe what each event represents and find their probability:

a. $A_1^c \cap A_2^c$

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b. How would you represent the event where they are awarded at least two projects? Find this probability.

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Describe what each event represents and find their probability:

c. How would you represent the event where they are awarded at least one project? Find this probability.

To-Do For Next Week

- [ ] Ensure you’re enrolled in a tutorial section
- [ ] Read and familiarize yourself with the course syllabus
- [ ] Review course learning outcomes
- [ ] Get a head start on textbook practice problems!

Have a great first week back!